



Cyber Security Project — 1st Deliverable

North African Hotel Pricing and Satisfaction Monitoring using Web
Scraping

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Contents

1	Introduction	2
2	Main Concepts and Theoretical Background	2
2.1	Definition of Web Scraping	2
2.2	Functional Flow	2
2.3	Robots.txt Protocol and Legal/Ethical Standards	3
3	Dataset Overview	3
4	Overview of Existing Solutions	3
4.1	Comparison of Web Scraping Tools	3
4.2	Detailed Analysis	4
5	Conclusion	4
6	References	4

1 Introduction

This project focuses on developing a web scraper for hotel price and customer satisfaction analysis on Booking.com. Booking.com is a global online travel platform that allows users to search, compare, and book accommodations, flights, and travel services.

The project covers five North African countries: Tunisia, Morocco, Algeria, Libya, and Mauritania. Data is collected across three seasonal periods to analyze pricing trends and customer satisfaction.

Web scraping is essential today to:

- Gather customer feedback
- Monitor competition and pricing trends
- Make data-informed decisions
- Conduct regional and seasonal market analysis

2 Main Concepts and Theoretical Background

2.1 Definition of Web Scraping

Web scraping is the process of automatically extracting data from websites using scripts or tools. The extracted data is typically transformed into structured formats such as CSV, Excel, or databases.

2.2 Functional Flow

The scraping process follows these steps:

1. Identify target website and search pages
2. Analyze HTML structure
3. Develop Selenium-based scraper
4. Iterate across countries, cities, and seasons
5. Store data in CSV format
6. Clean and preprocess data
7. Analyze pricing and satisfaction trends

2.3 Robots.txt Protocol and Legal/Ethical Standards

Robots.txt Protocol:

- Respect robots.txt rules
- Avoid server overload
- Do not extract sensitive data

Legal/Ethical Standards:

- Respect Terms of Service
- Avoid legal violations
- Cite data sources properly

3 Dataset Overview

This project builds a dataset from Booking.com including:

- Country and city
- Season and dates
- Hotel name and price
- Review score and label
- Review count
- Property features

The dataset will be used for analysis and visualization (e.g., Power BI) to study tourism trends and hotel performance.

4 Overview of Existing Solutions

4.1 Comparison of Web Scraping Tools

Tool	Description	Strengths	Limitations
BeautifulSoup	HTML parser	Easy to use, flexible	Slow for large-scale
Scrapy	Full framework	Fast, scalable	Steep learning curve

Selenium	Browser automation	Handles JavaScript	Resource intensive
Puppeteer	JS headless browser	Excellent JS support	JS only

4.2 Detailed Analysis

Solution	Advantages	Limitations
Scrapy	Customizable, scalable, pipeline support	Requires Python, complex
WebHarvy	No coding, automatic detection	Windows-only, limited features
Octoparse	User-friendly, cloud-based	Less flexible, paid features
Apify	Scalable, ready-made tools	Pricing, needs skills
Diffbot	AI extraction, no coding	Expensive, limited control

5 Conclusion

This project provides a structured approach to analyzing hotel pricing and customer satisfaction using web scraping.

Key highlights:

- Comparison of scraping tools
- Ethical scraping practices
- Dataset covering multiple countries and seasons
- Insights into pricing and review trends

Future work includes statistical analysis and sentiment classification.

6 References

1. Web Scraping with Python by Ryan Mitchell
2. <https://www.booking.com>
3. <https://developer.mozilla.org/en-US/>